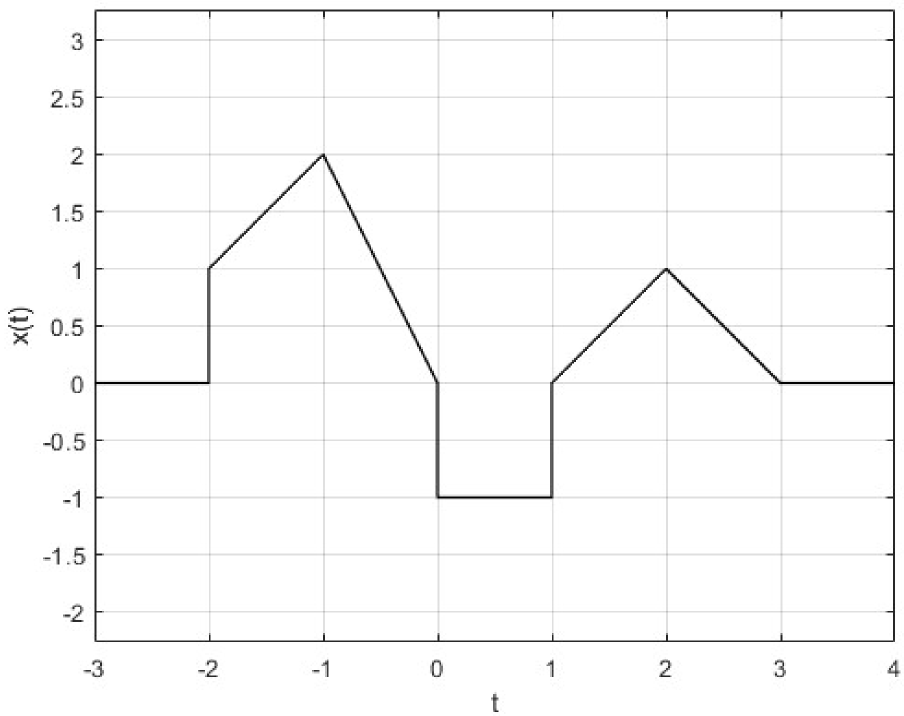
Lab Exercise 2

Independent Variable Manipulation using MatLab

Target Course Outcome:
CO1: Analyze signals and systems through an understanding of programming

Defining Signals

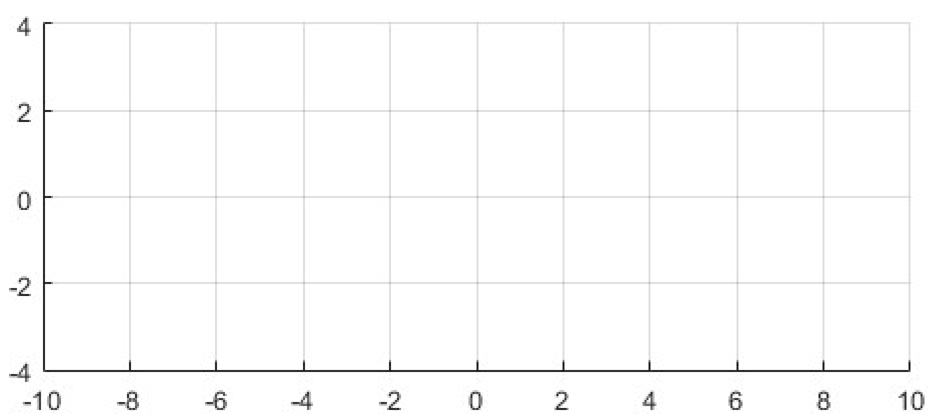
1. Given the signal below,



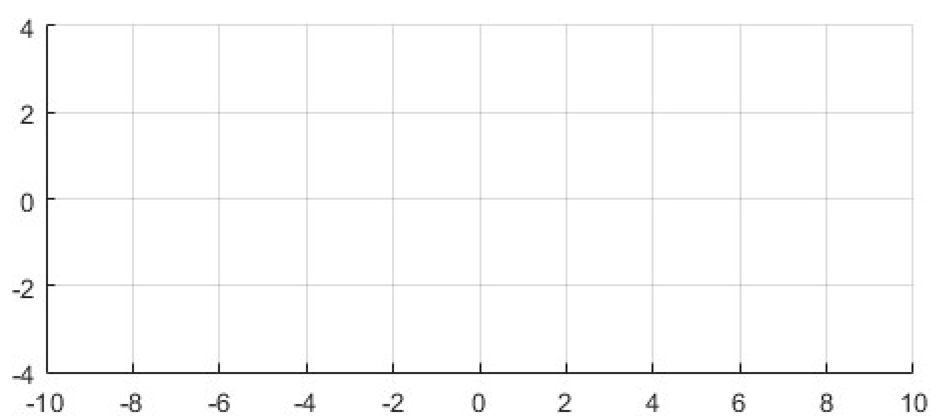
write down the piecewise definition (leave excess blanks unfilled):

$$x(t) = \begin{cases} \underline{\hspace{3cm}}, & \underline{\hspace{3cm}} \\ \underline{\hspace{3cm}}, & \underline{\hspace{3cm}} \\ \underline{\hspace{3cm}}, & \underline{\hspace{3cm}} \\ \underline{\hspace{3cm}}, & \underline{\hspace{3cm}} \\ \underline{\hspace{3cm}}, & \underline{\hspace{3cm}} \\ \underline{\hspace{3cm}}, & \underline{\hspace{3cm}} \\ \underline{\hspace{3cm}}, & \underline{\hspace{3cm}} \\ \underline{\hspace{3cm}}, & \underline{\hspace{3cm}} \\ \underline{\hspace{3cm}}, & \underline{\hspace{3cm}} \\ \underline{\hspace{3cm}}, & \underline{\hspace{3cm}} \end{cases}$$

2. Find
    a. x(-t+2) using the table below and plot its graph



b. x(-(t-1)/2) using the table below and plot its graph



3. Open MatLab
    a. Create a folder named LE2.
    b. Write a MatLab function that has the prototype

       function x = xt(t)

       and returns the value of the signal x(t) from number 1.
    c. Create a script file named “run_xt.m” and encode the source code below

```
t = -5 : 0.001 : 5; % you may have to adjust the start and end value of t
x = zeros(size(t));

for i = 1 : 1 : length(x)

  x(i) = xt( t(i) );

end

plot(t,x);
axis equal;
```

4. Modify “run_xt.m” naming it “run_xt_a.m” such that it now plots the signal x(-t+2) from 2a. Write your code below.

5. Modify “run_xt.m” naming it “run_xt_b.m” such that it now plots the signal x(-(t-1)/2) from 2b. Write your code below.

6. Given the discrete time signal,

   y[n] = {-3, -1, -1, 0, 1, 1, 2, 2, -1,1}
   ↑
   Recall that the arrow means the zero position

   a. Plot the signal using the stem function.

   b. Find z[n] = y[n/2]. Note that manipulation of the independent variable n must evaluate to an integer so use a “round down” or “floor” function to yield integer results.

n	**y[n]**	**n/2**	**z[n] = y[n/2]**